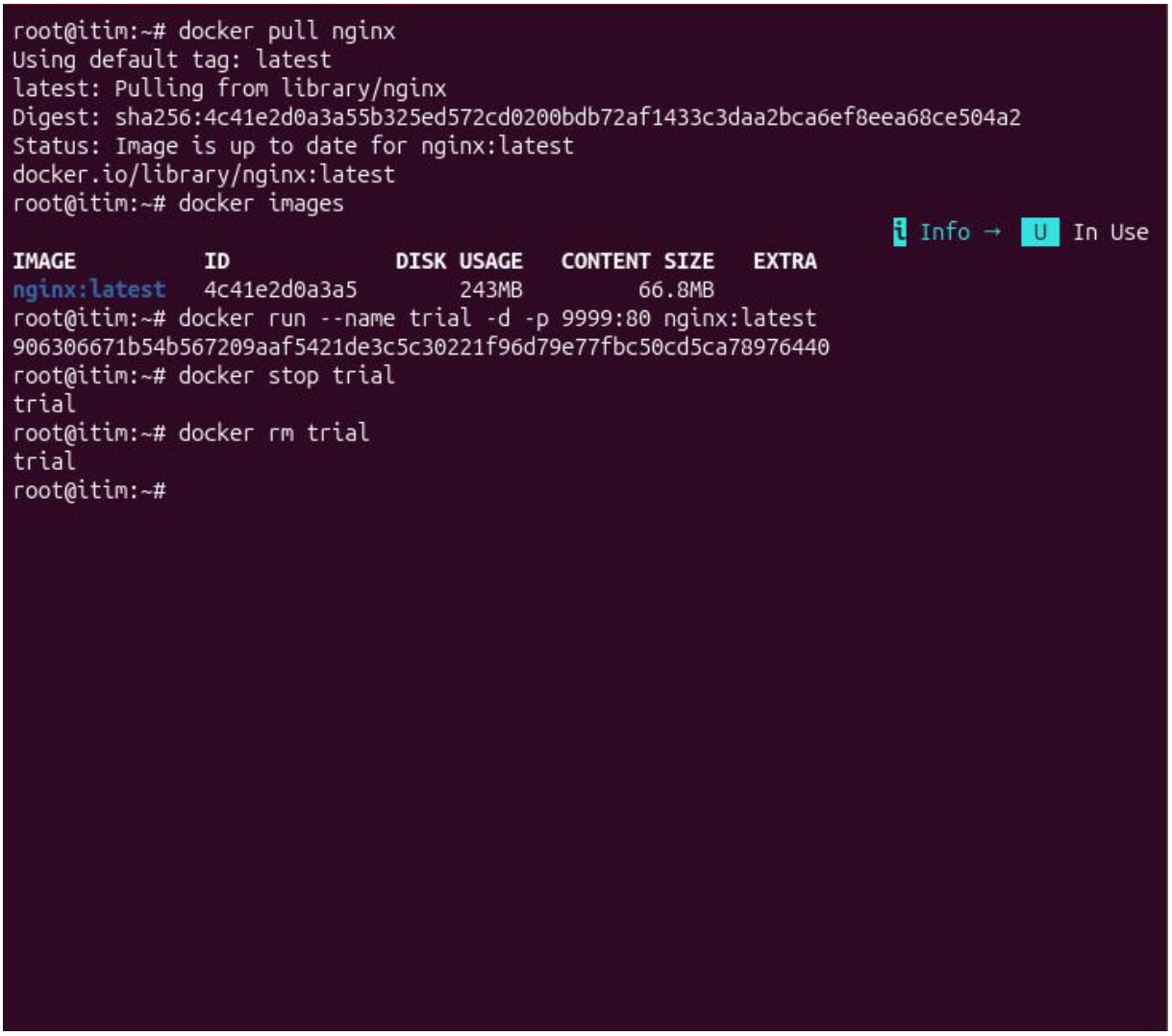
Assignment 2

1: Pull and Run Image

1. Pull the `nginx` image from Docker Hub.
2. Run a container from the image.
3. Check running containers.
4. Stop the container.
5. Remove the container.



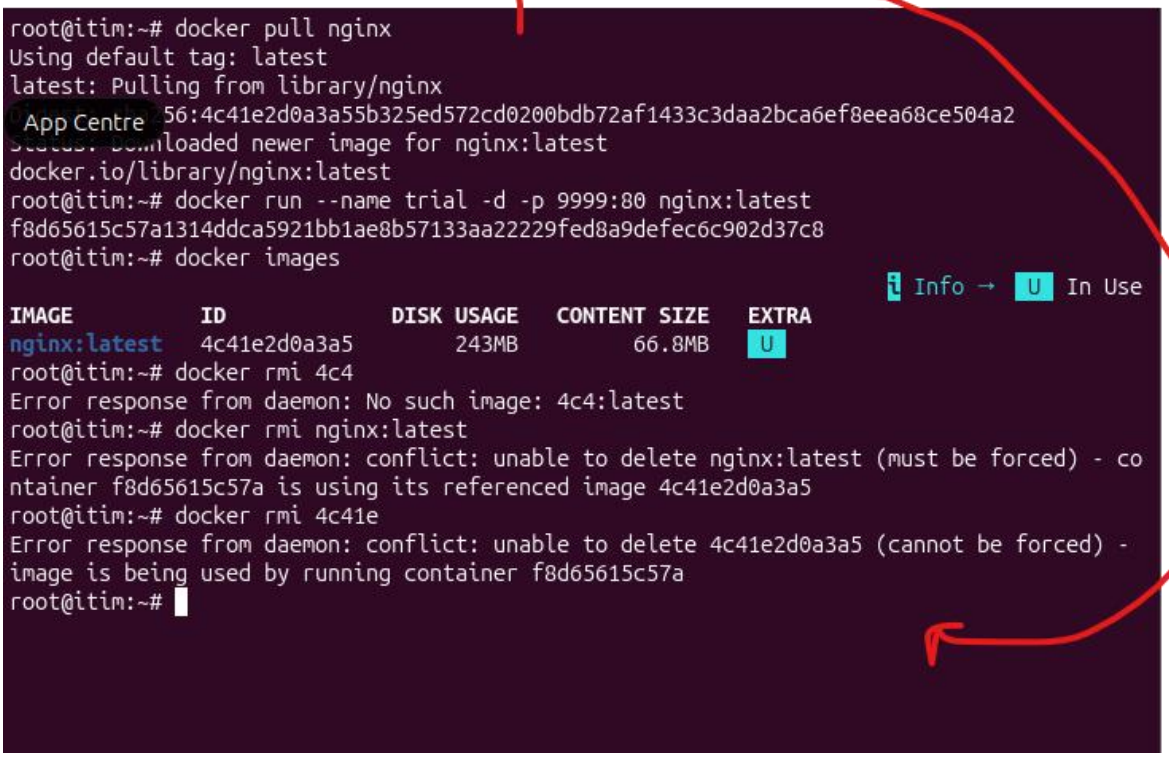
2: Image Commands Practice

1. Pull `alpine` image.
2. List all images available locally.
3. Inspect the `alpine` image.
4. Remove the `alpine` image.
5. What happens if you try to remove a running container's image? Explain.

```
root@itim:~# docker pull alpine:latest
latest: Pulling from library/alpine
7b229517ed22: Pull complete
3da1513d2124: Download complete
3ec36f3670df: Download complete
Digest: sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
root@itim:~# docker images
                                                    i Info →  U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest   be2656061b53         13MB         3.94MB
root@itim:~# docker inspect alpine
[
    {
        "Id": "sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a",
        "RepoTags": [
            "alpine:latest"
        ],
```

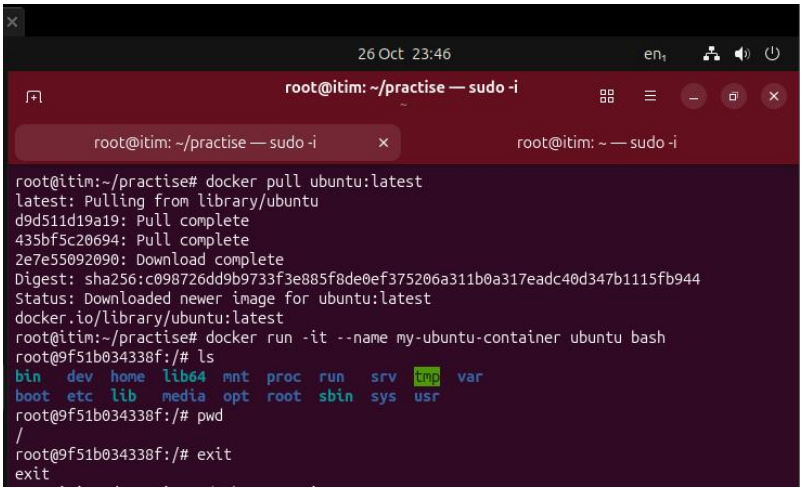
```
root@itim:~# docker images
                                                    i Info →  U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest   be2656061b53         13MB         3.94MB
root@itim:~# docker rmi be265
Untagged: alpine:latest
Deleted: sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a
root@itim:~#
```

Q 2 5th


root@itim:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
App Centre
56:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef8eea68ce504a2
loaded newer image for nginx:latest
docker.io/library/nginx:latest
root@itim:~# docker run --name trial -d -p 9999:80 nginx:latest
f8d65615c57a1314ddca5921bb1ae8b57133aa22229fed8a9defec6c902d37c8
root@itim:~# docker images
i Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
nginx:latest 4c41e2d0a3a5 243MB 66.8MB U
root@itim:~# docker rmi 4c4
Error response from daemon: No such image: 4c4:latest
root@itim:~# docker rmi nginx:latest
Error response from daemon: conflict: unable to delete nginx:latest (must be forced) - co
ntainer f8d65615c57a is using its referenced image 4c41e2d0a3a5
root@itim:~# docker rmi 4c41e
Error response from daemon: conflict: unable to delete 4c41e2d0a3a5 (cannot be forced) -
image is being used by running container f8d65615c57a
root@itim:~#

3: Container Commands

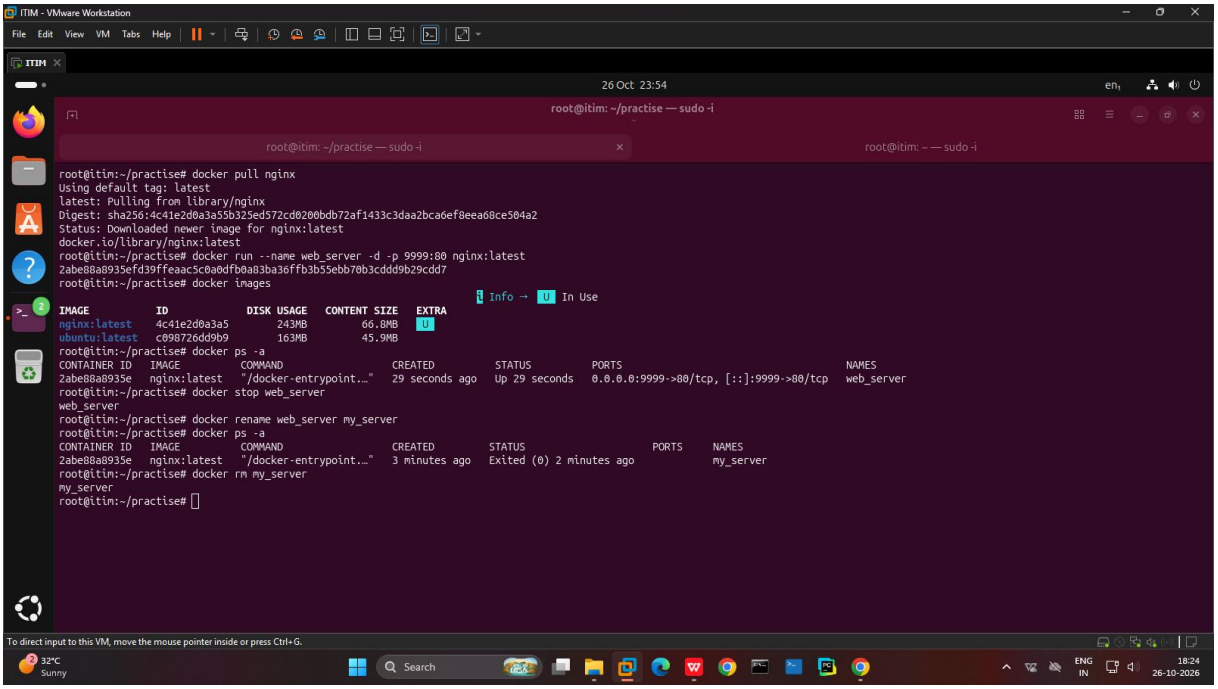
1. Run a container using `ubuntu` image in interactive mode.
2. Execute basic Linux command inside container (`ls` , `pwd`).
3. Exit the container.
4. List all containers (including stopped).
5. Restart the stopped container.





4: Naming and Renaming

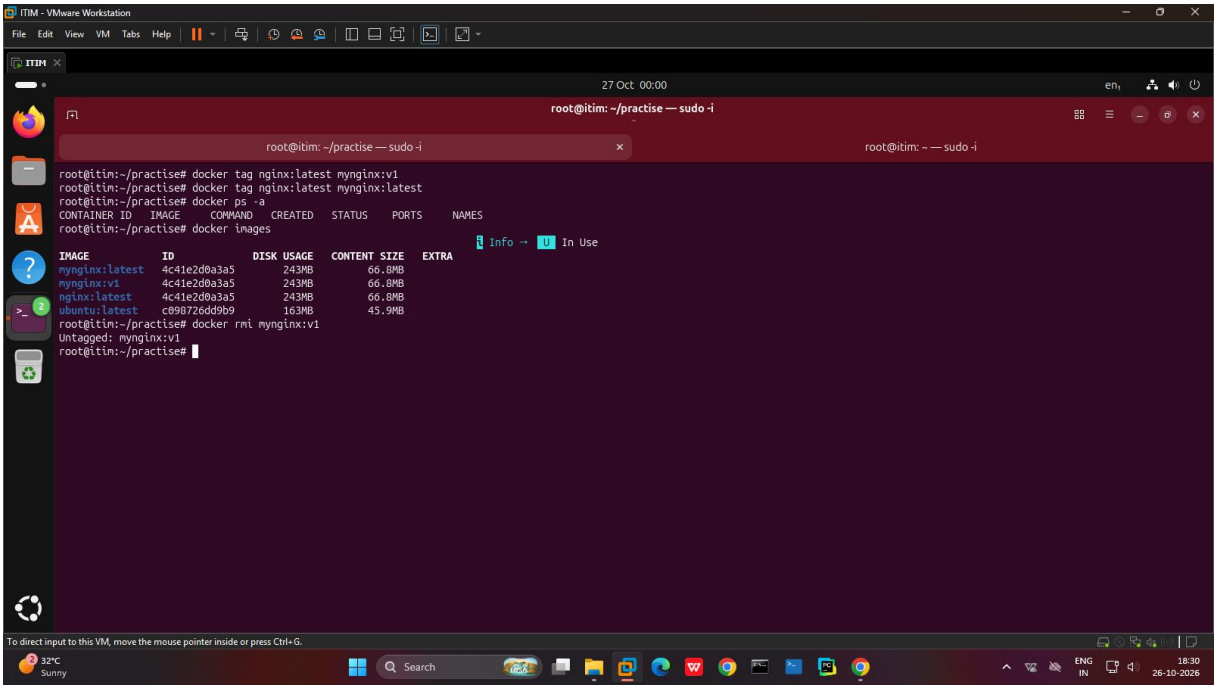
1. Run an `nginx` container with name `web_server` .
2. Verify the container name.
3. Stop the container.


ITIM - VMware Workstation
File Edit View VM Tabs Help
ITIM
26 Oct 23:54
root@itim: ~/practise — sudo -i
root@itim: ~/practise — sudo -i
root@itim: ~ — sudo -i
root@itim:~/practise# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef0eea68ce504a2
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
root@itim:~/practise# docker run --name web_server -d -p 9999:80 nginx:latest
2abe88a8935efd39ffeaac5c0a0dfb0a83ba36ffb3b55ebb70b3cddd9b29cdd7
root@itim:~/practise# docker images
I Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
nginx:latest 4c41e2d0a3a5 243MB 66.8MB U
ubuntu:latest c09872 6dd9b9 163MB 45.9MB
root@itim:~/practise# docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
2abe88a8935e nginx:latest "/docker-entrypoint.…" 29 seconds ago Up 29 seconds 0.0.0.0:9999->80/tcp, [::]:9999->80/tcp web_server
root@itim:~/practise# docker stop web_server
web_server
root@itim:~/practise# docker rename web_server my_server
root@itim:~/practise# docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
2abe88a8935e nginx:latest "/docker-entrypoint.…" 3 minutes ago Exited (0) 2 minutes ago my_server
root@itim:~/practise# docker rm my_server
my_server
root@itim:~/practise#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
32°C Sunny
Search
ENG IN
18:24 26-10-2026

4. Rename the container to `my_server`.
5. Remove the container.

5: Tagging Docker Image

1. Pull `nginx` image.
2. Tag it as `mynginx:v1`.
3. Tag it again as `mynginx:latest`.
4. List images and verify tags.
5. Remove one tag and observe result.

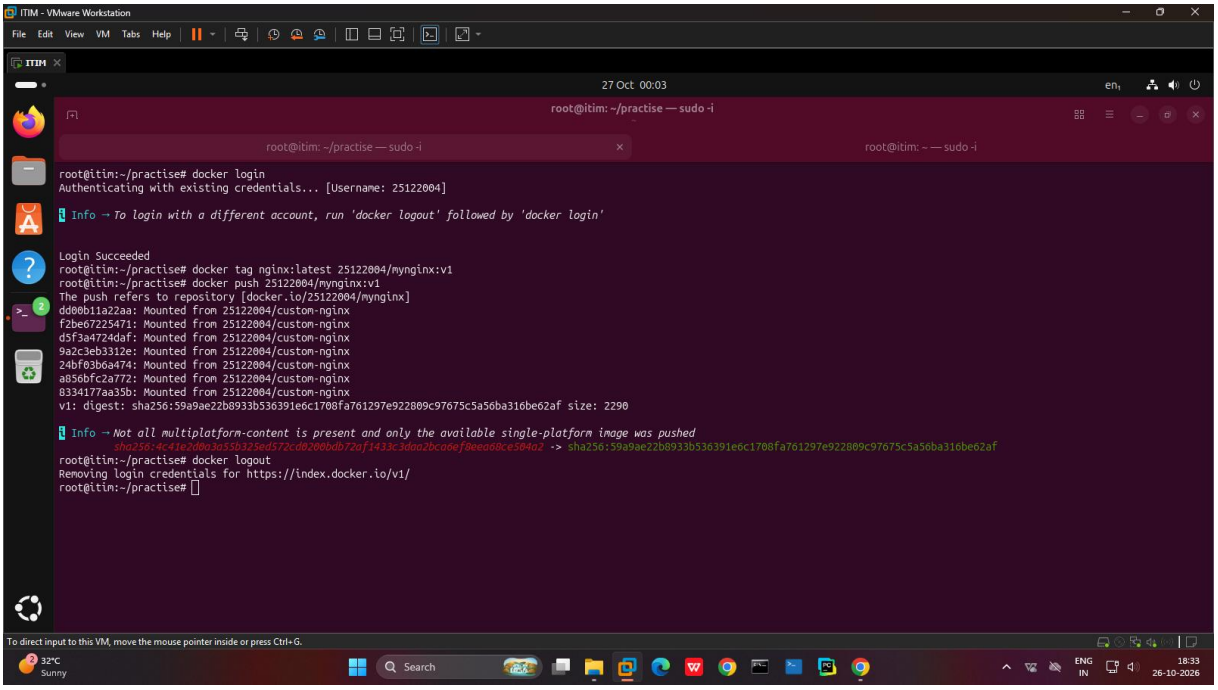


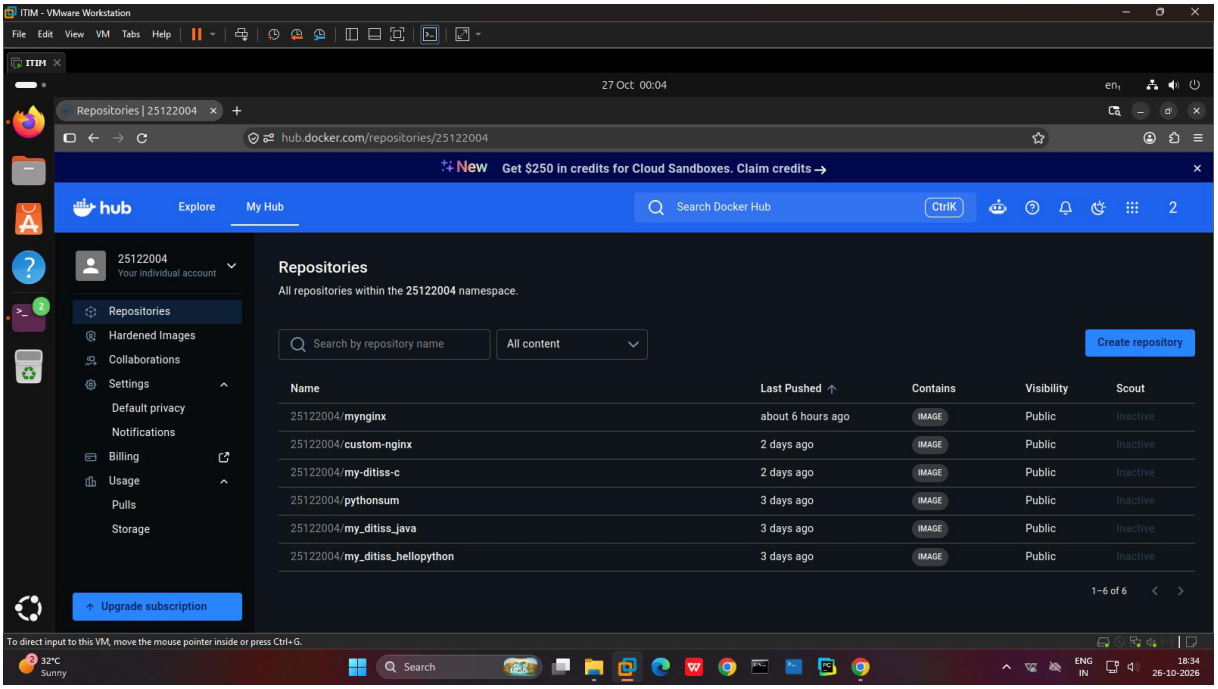
6: Push Image to Docker Hub

1. Login to Docker Hub using `docker login`.
2. Tag your image as:

```
<your_username>/mynginx:v1
```

3. Push the image to Docker Hub.
4. Logout from Docker.
5. Verify image on Docker Hub website.




ITIM - VMware Workstation
File Edit View VM Tabs Help
27 Oct 00:04
Repositories | 25122004
hub.docker.com/repositories/25122004
New Get $250 in credits for Cloud Sandboxes. Claim credits
hub
Explore
My Hub
Search Docker Hub
CtrlK
25122004
Your individual account
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade subscription
Repositories
All repositories within the 25122004 namespace.
Search by repository name
All content
Create repository
Name
Last Pushed
Contains
Visibility
Scout
25122004/mynginx
about 6 hours ago
IMAGE
Public
Inactive
25122004/custom-nginx
2 days ago
IMAGE
Public
Inactive
25122004/my-ditiss-c
2 days ago
IMAGE
Public
Inactive
25122004/pythonsum
3 days ago
IMAGE
Public
Inactive
25122004/my_ditiss_java
3 days ago
IMAGE
Public
Inactive
25122004/my_ditiss_hellopython
3 days ago
IMAGE
Public
Inactive
1–6 of 6
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
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18:34
26-10-2026

7: Basic Networking and Port Mapping

1. Run an `nginx` container with port mapping:

```
8080:80
```

2. Open browser and access `localhost:8080`.
3. List Docker networks.
4. Inspect default bridge network.
5. Explain what port mapping does in Docker.

